Supplementary Texts (Texts S1-S3)

Supplementary Text S1. *Eligibility Criteria*

Inclusion

- Language restriction to English, Spanish, French, and Russian.
- Observational studies (Case control, cohort, cross-sectional, nested designs).
- Full text availability.
- Patients diagnosed with breast cancer (any stage).
- Patients treated with taxanes (Paclitaxel, docetaxel).
- Adverse effects assessment.
- Neurotoxicity assessment.
- Genetic variation assessments (either single nucleotide polymorphisms, copy number variation, short tandem repeats/microsatellite variation, restriction fragment length polymorphisms, small-scale insertions/deletions).
- Prevalence of minor alleles by phenotype in tabular or redacted form.

Exclusion

- Duplicated results from sources of information (Ovid, PubMed, Scopus, Web of Science, Virtual Health Library/LILACS, ProQuest, and OpenGrey).
- No full text availability.
- Abstracts or conference proceedings without full-text availability.
- Full text in languages other than English, Spanish, French, and Russian.
- Prior narrative or systematic reviews on neurotoxicity and taxanes.
- Single or family case reports.
- Clinical research in breast cancer, taxane toxicity, or neurotoxicity alone other than genetic variation.
- Basic research in breast cancer treatment toxicities: animal studies or cell cultures, computational methods.
- Other clinical or basic reports in systemic diseases.

Supplementary Text S2. *Systematic Search Protocol.*

RESEARCH QUESTION

Highlighted in bold "hot" or key terms in structured review question

Population: females, breast cancer, all subtypes, all stages, therapy with taxanes (paclitaxel, docetaxel)

Exposition: Genomic markers (including SNPs, miRNAs, CNVs, etc.) (Minor allele carriers considered as exposed) in any gene.

Comparison: Major allele carriers (taken as unexposed).

Outcome: Neurotoxicity (Peripheral neuropathy, Central Neurotoxicity)

Among females diagnosed with **breast cancer** (all subtypes and stages) under treatment with **taxanes (paclitaxel, docetaxel)**, does any **genomic marker** (e.g., SNP, CNV, indel, miRNA) significantly influence the **probability/risk** of developing **neurotoxicity**? Additionally, does any genomic marker is predominantly associated with **peripheral neuropathy or central neurotoxicity**?

SEARCH STRATEGY

Systematic searches

1) **Ovid/Ovid Medline/Embase Search (Date of search: 7/6/20 2:37 PM) N = 738 with duplicates; 601 without duplicates**
   Filtered by Type of Work and selecting: Article; Journal Article; Multicenter Study; Comparative Study; Observational study; Research Support/Report N = 205

Hot/Key terms	**Breast cancer**	**Genomic markers (SNP, CNV, indel, STRP, STR, RFLP) Genetic variation**	**Taxanes (Paclitaxel, docetaxel)**	**Neurotoxicity**
Controlled vocabulary OR Indexed terms	exp breast neoplasms/ or exp breast carcinoma in situ/ or exp breast neoplasms, male/ or exp carcinoma, ductal, breast/ or exp carcinoma, lobular/ or exp "hereditary breast and ovarian cancer syndrome"/ or exp inflammatory breast neoplasms/ or exp triple negative breast neoplasms/ or exp unilateral breast neoplasms/	exp genetic linkage/ or exp linkage disequilibrium/ or exp lod score/ or exp genetic load/ or genetic variation/ or exp genetic heterogeneity/ or exp mutation/ or exp polymorphism, genetic/ or exp genomic structural variation/ or exp pharmacogenomic variants/ or exp polymorphism, restriction fragment length/ or exp polymorphism, single nucleotide/ or exp polymorphism, single-stranded conformational/ or exp genotype/ or exp gene dosage/ or exp genetic predisposition to disease/ or exp haplotypes/ or exp hemizygote/ or exp heterozygote/ or exp homozygote/ or exp inheritance patterns/	exp taxoids/ or exp docetaxel/ or exp paclitaxel/	–

N	838043	3286247	184442	0
Boolean operators/ Textword search strings	(breast adj1 cancer).ab,kw,ti,tw,tx. (breast adj1 tumor).ab,kw,ti,tw,tx. (breast adj1 carcinoma*).ab,kw,ti,tw,tx. (breast adj1 malignanc*).ab,kw,ti,tw,tx.	"polymorphism*".ab,kw,ti,tw,tx.	"taxane*".ab,kw,ti,tw,tx. "taxoid*".ab,kw,ti,tw,tx. paclitaxel.ab,kw,ti,tw,tx. docetaxel.ab,kw,ti,tw,tx. (nab adj1 paclitaxel).ab,kw,ti,tw,tx. (albumin adj3 paclitaxel).ab,kw,ti,tw,tx.	NTX.ab,kw,ti,tw,tx. "neurotoxicit*".ab,kw,ti,tw,tx. "neuropath*".ab,kw,ti,tw,tx. "peripheral neuropath*".ab,kw,ti,tw,tx. TIPN.ab,kw,ti,tw,tx. "central neuropath*".ab,kw,ti,tw,tx. "central neurotoxicit*".ab,kw,ti,tw,tx. CIPN.ab,kw,ti,tw,tx. (taxane adj3 neuropath*).ab,kw,ti,tw,tx. (chemotherap* adj3 neuropath*).ab,kw,ti,tw,tx. (paclitaxel adj1 acute adj1 pain adj1 syndrome).ab,kw,ti,tw,tx.
N	1062685	916154	219573	759915
Other				
N				
TOTAL	1310270	4202401	301208	759915

2) **Pubmed Search (Date of search: 7/6/20 1:45 PM) N = 37**
(("Genetic Variation"[Mesh]) OR (polymorphism[Title/Abstract)) → 1,139,986 results
AND
(("Breast Neoplasms"[Mesh]) OR (breast cancer[Title/Abstract]) OR (breast neoplasm[Title/Abstract]) OR (breast carcinoma[Title/Abstract])) → 376,801results
AND
(("Taxoids"[Mesh]) OR (paclitaxel[Title/Abstract]) OR (docetaxel[Title/Abstract])) → 51,598 results
AND
(((neuropathy[Title/Abstract]) OR (peripheral neuropathy[Title/Abstract])) OR (neurotoxicity[Title/Abstract])) → 103,644 results

3) **ProQuest Search (Date of search: 7/6/20 11:05 PM) N = 17**

tiab(breast cancer OR breast carcinoma OR breast neoplasm) AND tiab(polymorphism OR genotype OR pharmacogenetic OR haplotype) AND tiab(paclitaxel OR taxane OR taxoid OR docetaxel) AND tiab(neuropathy OR neurotoxicity OR peripheral neuropathy OR central neurotoxicity)

4) **Scopus Search (Date of search: 7/6/20 10:41 PM) N = 196**
Filtered to Article/Editorial/Letter N = 101
TITLE-ABS-KEY(polymorphism*) OR TITLE-ABS-KEY(SNP*) OR TITLE-ABS-KEY(genotyp*) OR TITLE-ABS-KEY(haplotyp*) OR TITLE-ABS-KEY(pharmacogen*)
AND
TITLE-ABS-KEY(breast cancer*) OR TITLE-ABS-KEY(breast carcinoma*) OR TITLE-ABS-KEY(breast neoplasm*)
AND
TITLE-ABS-KEY(Taxoid*) OR TITLE-ABS-KEY(Taxane*) OR TITLE-ABS-KEY(paclitaxel) OR TITLE-ABS-KEY(docetaxel)
AND
TITLE-ABS-KEY(neuropath*) OR TITLE-ABS-KEY(peripheral neuropath*) OR TITLE-ABS-KEY(central neurotoxicit*) OR TITLE-ABS-KEY(neurotoxicit*)

5) **Virtual Health Library (Date of search: 7/6/20 10:51 PM) N = 6**
(tw: genetic variation OR genetic marker OR polymorphism)
AND
(tw: breast cancer OR breast carcinoma OR breast neoplasm)
AND
(tw: taxoid OR taxanes OR paclitaxel OR docetaxel)
AND
(tw: neurotoxicity OR peripheral neuropathy OR central neurotoxicity)

6) **Web of Science (Date of search: 7/6/20 10:41 PM) N = 75**
Refined by: DOCUMENT TYPES: (ARTICLE OR EARLY ACCESS OR CORRECTION OR LETTER)
Timespan: All years. Indexes: SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI. N = 61
ALL=(polymorphism* OR snp* OR genetic marker* OR genotyp* OR haplotype* OR pharmacogen*)
AND
ALL=(breast cancer* OR breast carcinoma* OR breast neoplasm*)
AND
ALL=(taxoid* OR taxane* OR paclitaxel OR docetaxel)
AND
ALL=(neuropath* OR peripheral neuropath* OR central neurotoxicit* OR neurotoxicit*)

Supplementary Text S3. *Risk of bias profile*

GRADED BIASES

Valid phenotype, ***Neurotoxicity***	Clear **description** or definition consistently used in the manuscript. Phenotype assessed by experts (Oncologists, Peripheral nerve specialists – neurology, plastic surgery, orthopedics, podologists). **Uses the following scales: NCI-CTC OR TNS.**
Valid phenotype, ***Breast cancer***	Diagnosis supported by international guidelines, preferably ASCO/ESMO/JNCCN.
Observer Bias	**Blinded neurotoxicity assessors.**
Recall bias	Whether the assessments were done from other sources other than the **electronic medical records**. **Less optimal self-report.**
Valid genotyping	Genotyping quality control. Results done by **duplicate (also by different method), blinded experienced lab technicians, SNP <0.01, genotyping call rate >95%, SNP call rate >95%**, report of providers of reagents e.g.ThermoFisher scientific, etc.
Stage migration	Neurotoxicity. Combining patients across scales (NCI-CTC + TNS + self-report)
Stage migration	Breast cancer. Analyzing patients across TNM stages or disease stages (Early (1-2a), Locally advanced (2b – 3c), Metastasic/Advanced (4)).
Multiple publication	Author with the **same publication in two different journals.**
Attrition bias	**Loss >20% of sample per cancer stage or total.**
Selective outcome reporting	Only a subset of the original outcomes measured and analyzed in a study are fully reported based on the magnitude of the treatment effect or the statistical significance of selected outcomes. **Only a subset of the sample is described without reasons in text.**
Multiple comparison	Whether the authors' performed p-value correction. A typical correction is to use a new p-value calculated by: 0.05/number associations examined. **Bonferroni, FDR, Benjamini-Hochberg**
Population stratification	Genetic differences between cases and controls unrelated to disease but due to sampling from populations of different ancestries. **Only ONE race/ethnicity OR analysis are done SEPARATE**
Hardy-Weinberg Equilibrium	Whether controls in cohort/case-control/cross-sectional designs were in **Hardy-Weinberg equilibrium (p>0.05)** or if the authors prove there were no founder effects or if allele frequencies are reported.
Survivor bias	If the paper analyzed stage 4 or locally advanced, **mortality and time to assessment of neuropathy should be equal.**
Mendelian randomization	**SNP or gene is known to be associated with neuropathy AND breast cancer.**

Every bias was graded as either HIGH, INTERMEDIATE or LOW risk
HIGH risk → The paper clearly states a bias.
INTERMEDIATE risk → Unclear from reading paper.
LOW risk → Evidence on the paper that there is NO bias.